Business Requirements Document (BRD)

Merchant Intelligence Platform for Cards Business

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1. Executive Summary

Project Name

Merchant Intelligence Platform (MIP) for Cards Business

Business Vision

Create an enterprise-wide Merchant Intelligence Platform that transforms raw card transaction data into actionable merchant insights, enabling banks to identify merchant growth opportunities, improve acquiring business performance, optimize merchant acquisition strategies, increase card spend, and generate new revenue streams.

Business Problem

Banks process billions of card transactions across merchants but face several challenges:

- Limited visibility into merchant ecosystem dynamics
- Lack of merchant benchmarking capabilities
- Inability to identify high-growth merchant opportunities
- Difficulty measuring merchant wallet share
- Absence of merchant segmentation intelligence
- Limited understanding of customer spending behavior across merchants
- Delayed identification of merchant acquisition opportunities
- Lack of predictive insights for merchant growth

As a result:

- Merchant acquisition efforts are inefficient
- Cross-sell opportunities are missed
- Merchant retention rates decline
- Acquiring revenues are not optimized
- Marketing campaigns lack merchant-level intelligence

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2. Business Objectives

Objective	Description
Increase Acquiring Revenue	Grow merchant acquiring business by identifying high-potential merchants
Improve Merchant Retention	Detect churn risks proactively
Merchant Cross-Sell Growth	Recommend merchant-focused banking products
Merchant Benchmarking	Compare merchants against peers
Market Share Expansion	Identify wallet-share capture opportunities
Digital Adoption Growth	Increase card payment penetration
Merchant Acquisition Optimization	Focus acquisition efforts on targeted merchants
Customer Engagement Enhancement	Improve merchant offers and rewards

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3. Business Stakeholders

Primary Stakeholders

Cards Business Head

Responsible for acquiring business growth.

Merchant Acquiring Team

Responsible for merchant onboarding and retention.

Payments Strategy Team

Responsible for market expansion.

Product Management Team

Responsible for merchant products and pricing.

Relationship Managers

Manage merchant relationships.

Secondary Stakeholders

- Retail Banking
- Corporate Banking
- Risk Management
- Data Analytics Team
- Marketing Team
- Finance Team

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4. Scope

In Scope

Merchant Analytics

- Merchant performance tracking
- Merchant segmentation
- Merchant clustering
- Growth analytics
- Merchant lifecycle analysis

Customer Spend Analytics

- Spend patterns
- Merchant affinity
- Cross-shopping insights

Geographic Analytics

- Merchant density analysis
- Geo-spatial trend analysis
- Regional performance insights

Competitive Intelligence

- Wallet share estimation
- Merchant benchmarking
- Competitor analysis

Predictive Intelligence

- Growth prediction
- Churn prediction
- Spend forecasting

AI Insights

- Merchant opportunity scoring
- Next best action recommendations
- Market basket analysis

Out of Scope

- Merchant payment gateway processing
- Real-time authorization systems
- Fraud detection systems
- Settlement systems
- Chargeback management

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5. Current State Assessment

Existing Process

Banks currently:

1. Receive transaction feeds
2. Produce static merchant reports
3. Perform manual analysis
4. Use siloed merchant systems
5. Lack predictive insights

Current Issues

Issue	Impact
Data Silos	Incomplete merchant view
Static Reporting	Delayed action
Limited Segmentation	Poor targeting
No Predictive Models	Lost revenue opportunities
No Market Insights	Reduced competitiveness

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6. Future State Vision

The Merchant Intelligence Platform will provide:

Single Merchant 360 View

Comprehensive profile including:

- Merchant demographics
- Transaction history
- Revenue metrics
- Customer segments
- Geographic performance
- Growth opportunities

AI Powered Insights

Automated recommendations:

- Merchant acquisition targets
- Churn prevention actions
- Product cross-sell opportunities
- Revenue improvement opportunities

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7. Business Requirements

BR-01 Merchant 360 Intelligence

Requirement

System shall provide a comprehensive merchant profile.

Data Elements

- Merchant ID
- Merchant Name
- Merchant Category
- Industry Classification
- Merchant Location
- Merchant Size
- Onboarding Date
- Revenue Metrics
- Customer Metrics

BR-02 Merchant Segmentation

Requirement

Automatically classify merchants into segments.

Segments

Revenue Based

- Platinum
- Gold
- Silver
- Bronze

Business Size

- Enterprise
- Mid-Market
- SME
- Micro

Growth Stage

- Emerging
- Growth
- Mature
- Declining

BR-03 Merchant Performance Dashboard

System shall provide KPIs.

Metrics

- Total Sales
- Total Transaction Value
- Transaction Volume
- Average Ticket Size
- Revenue Growth %
- Active Customers
- Repeat Customers
- Merchant Share

BR-04 Merchant Benchmarking

Compare merchant against peer group.

Benchmark Metrics

- Spend Growth
- Transaction Growth
- Average Ticket Size
- Customer Loyalty
- Visit Frequency
- Regional Rank

BR-05 Wallet Share Analysis

Estimate merchant wallet share.

Outputs

- Current Wallet Share
- Potential Wallet Share
- Wallet Leakage
- Competitor Wallet Share

BR-06 Merchant Network Analytics

Discover merchant ecosystem patterns.

Analysis

- Merchant Affinity
- Customer Migration
- Cross Shopping Behavior
- Merchant Clusters

BR-07 Geo-Spatial Intelligence

Map merchant activities geographically.

Features

- Heatmaps
- Growth Zones
- White Space Opportunities
- Market Penetration Analysis

BR-08 Merchant Growth Opportunity Engine

Identify potential acquisition opportunities.

Output

- Opportunity Score

- Expected Revenue
- Acquisition Priority
- Recommended Actions

BR-09 Merchant Churn Prediction

Predict merchant attrition.

Features

- Churn Score
- Churn Probability
- Root Cause Analysis
- Retention Action Suggestions

BR-10 Customer Spending Intelligence

Analyze consumer spending trends.

Metrics

- Spend Frequency
- Average Spending
- Basket Size
- Spend Categories
- Loyalty Metrics

BR-11 Campaign Effectiveness Analytics

Evaluate merchant campaigns.

Outputs

- Incremental Spend
- Redemption Rate
- ROI
- Customer Conversion

BR-12 AI Recommendation Engine

Generate intelligence-driven recommendations.

Recommendation Types

- Merchant Acquisition
- Merchant Retention
- Campaign Targeting
- Pricing Optimization
- Product Cross-Sell

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8. Functional Requirements

Merchant Data Management

FR-001 Merchant Master Integration

System shall integrate merchant master records from:

- Merchant Onboarding System
- CRM System
- Acquiring Platform

FR-002 Merchant Profile Creation

Automatically create merchant profiles.

FR-003 Merchant Enrichment

Enrich merchant records with:

- Industry Classification
- Geo Coordinates
- Regional Mapping

Analytics

FR-004 Merchant Segmentation Engine

Create dynamic merchant segments.

Frequency:

- Daily refresh

FR-005 Benchmark Analytics

Generate peer-group comparisons.

Frequency:

- Daily

FR-006 Opportunity Scoring

Calculate merchant opportunity score.

Frequency:

- Weekly

FR-007 Predictive Analytics

Run ML models for:

- Growth Prediction
- Churn Prediction
- Spend Forecasting

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9. Non-Functional Requirements

Performance

Requirement	Target
Dashboard Load	< 5 seconds
KPI Refresh	Daily
Query Response	< 3 seconds

Requirement	Target
Concurrent Users	5,000+

Scalability

Support:

- 1+ Billion Transactions
- 10+ Million Cards
- 5+ Million Merchants

Availability

- 99.9% uptime

Security

- PCI-DSS compliant
- Data Encryption
- RBAC
- Audit Logging

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10. Input Datasets

Internal Sources

Card Transactions

Attribute
Transaction ID
Merchant ID
Card Number Token
Amount

Attribute
Date
Time
MCC
Channel

Merchant Master

Attribute
Merchant ID
Merchant Name
MCC
Industry
Location
Acquisition Date

Customer Master

Attribute
Customer ID
Segment
Demographics
Geography

CRM Data

Attribute
Lead Status
Relationship Manager
Merchant Interactions

External Sources

- Census Data
- Merchant Directories
- Economic Indicators
- Industry Growth Data
- GIS Data

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11. KPI Matrix

KPI	Formula
Total Spend	SUM(Transaction Amount)
Merchant Revenue	MDR Revenue + Fees
Avg Ticket Size	Spend / Transactions
Growth %	(Current - Previous)/Previous
Repeat Rate	Repeat Customers / Total Customers
Wallet Share	Merchant Spend / Total Spend
Churn Rate	Lost Merchants / Total Merchants
Opportunity Score	ML Score

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12. Dashboard Requirements

Executive Dashboard

Widgets

- Merchant Count
- Total Spend
- Revenue Trend
- Growth Rate
- Churn Trend
- Opportunity Pipeline

Merchant Performance Dashboard

Widgets

- Merchant Ranking
- Spend Trend
- Transaction Trend
- Benchmark Comparison
- Geo Performance

Acquisition Dashboard

Widgets

- Opportunity Pipeline
- Prospect Ranking
- Market Coverage
- Merchant Density

Churn Dashboard

Widgets

- Merchant Risk Heatmap
- Churn Drivers
- Retention Opportunities

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13. AI/ML Capabilities

Model 1: Merchant Growth Prediction

Prediction

Forecast:

- Revenue Growth
- Transaction Growth
- Customer Growth

Model 2: Churn Prediction

Output

- Churn Probability
- Contributing Factors
- Risk Segment

Model 3: Merchant Clustering

Algorithms:

- K-Means
- DBSCAN

Outputs:

- Merchant Archetypes
- Peer Groups

Model 4: Recommendation Engine

Outputs:

- Next Best Merchant
- Next Best Product
- Next Best Offer

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14. Benefits & Expected Business Value

Benefit	Expected Impact
Merchant Revenue Growth	10%–20%
Merchant Acquisition Efficiency	25%–35%
Merchant Retention Improvement	15%–25%
Cross-Sell Increase	20%–30%
Campaign ROI Improvement	15%–20%
Market Share Growth	5%–10%

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15. Success Criteria

The solution shall be considered successful when:

- Merchant acquisition conversion improves by 25%
- Merchant churn reduces by 15%
- Acquiring revenue grows by 10%
- 90% of merchants are segmented automatically
- 80% of acquisition campaigns use platform insights
- Executive dashboards refresh daily
- AI recommendation accuracy exceeds 75%

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16. Implementation Roadmap

Phase 1 – Data Foundation (8 Weeks)

- Data ingestion
- Merchant master integration
- Transaction integration
- Data quality framework

Phase 2 – Merchant Intelligence Layer (10 Weeks)

- Merchant 360
- KPI computations
- Segmentation engine
- Benchmarking

Phase 3 – Advanced Analytics (8 Weeks)

- Churn prediction
- Growth forecasting
- Opportunity scoring

Phase 4 – AI & Optimization (6 Weeks)

- Recommendation engine
- Generative insights
- Automated decision support

Phase 5 – Enterprise Rollout (4 Weeks)

- UAT
- Training
- Production deployment

This BRD is suitable for transformation into a full Solution Design Document (SDD), including detailed source-to-target mappings, star schema data model, KPI calculation matrix, Azure architecture, data lineage, dashboard wireframes, and AI/ML architecture for enterprise implementation.